

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 21

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Silver carp eat phytoplankton, bighead carp zooplankton, grass carp and tilapia vegetation, common carp bottom-feed on molluscs / insects, catfish or perch eat small fish • Species occupy different niches / trophic levels so they do not compete and all food sources are used • Bottom feeders re-suspend nutrients boosting plankton / plant growth • Predators control over-breeding species (tilapia) so remaining fish grow larger; total yield exceeds any single species	4
01.2	• IMTA: fed species (salmon, shrimp) alongside inorganic extractive species (seaweeds absorbing ammonia and phosphate) and organic extractive species (filter-feeding mussels eating waste particles / plankton) • Wastes of one species become nutrients for another so less pollution and extra harvests • Aquaponics: aquaculture water rich in nutrients / organic matter feeds hydroponic salad crops (lettuce, herbs, watercress) • Plants remove the nutrients that would cause deoxygenation and cleaned water returns to the fish (tilapia, carp, catfish); rice-fish paddies similar	4
01.3	• Bluefin tuna do not breed well in captivity so young fish are caught from the wild and fattened in sea cages • Production still depends on (and removes juveniles from) the wild population; carnivores need large fish inputs	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 357-358; AQA spec 3.5.2.3.	
02.1	• The new opportunity is overwhelmingly attractive (economic benefit) • Lack of knowledge / understanding of the impacts • The impacts occur elsewhere or much later, so are not evident to those who benefit	3

Question	Marking guidance	Marks
02.2	• Cultural belief that resources should be exploited to satisfy needs only • Problems of over-exploitation become obvious quickly as local supplies dwindle, so behaviour changes	2
02.3	• Resources are brought in from other areas • If one area is depleted, exploitation moves to another area (eg timber, fish, metals, energy) without the consumers noticing	2
02.4	• Easier access to energy has made many new activities possible • The scale of activities has greatly increased • Toxic wastes are produced that cannot be decontaminated by natural processes	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-371; AQA spec 3.6.1.	
03.1	• Surveys monitor photosynthesis rates, biomass, soil water content and the spread of pests / diseases so inputs are applied only where needed • Yield mapping of different parts of a field controls fertiliser application in each area for future crops • This maximises yield while avoiding wastage / reducing leaching and nutrient pollution	3
03.2	• Machinery, pesticides, fertilisers, improved breeds and infrastructure (transport, refrigeration) are expensive • They are mainly available in affluent societies; poorer farmers cannot afford the inputs	2
03.3	• Provides an income that allows producers to afford basic rights such as water, education, health care and food	1
03.4	• Nitrates are leached into water bodies used as sources of potable water • High nitrate levels cause blue baby syndrome (methaemoglobinaemia) and nitrates may be carcinogenic	2
03.5	<i>Any 2 from:</i> • Anaerobic digestion by microbes in rice padi fields • Livestock intestines (ruminant digestion) • Anaerobic decay of manure / slurry	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-332; AQA spec 3.5.1.3-3.5.1.4.	
04.1	• Small clearings were farmed for a few years then abandoned as nutrients declined; the forest recolonised and soil fertility recovered before the area was cleared again • Population growth shortens the gap between clearances so the forest and soil do not recover • Clearings enlarge and degrade; less regrowth	3
04.2	• Rainforest soils are thin and nutrient-poor (nutrients held in the biomass); once cleared, fertility falls in a few years and yields drop, so more land is cleared • Abandoned land may not be recolonised if no forest survives nearby; erosion of exposed soil	2

Question	Marking guidance	Marks
04.3	• Mining (aluminium, iron, gold, copper) destroys habitat directly and pollutes rivers with turbid drainage, cyanide and mercury from processing • Reservoirs (eg Tukurui, 2850 km²) flood large areas, block migratory fish and river dolphins; decomposing vegetation produces acidic, deoxygenated water; fluctuating downstream flows • Droughts increase frequency, area and duration of fires (long fires boil sap and kill trees); cloud forests lose humidity as clouds form higher so frogs and epiphytes decline; some models predict replacement by savannah	3
04.4	• A creditor country or conservation charity cancels / buys part of a debtor country's foreign debt • In return the country guarantees to protect an area of rainforest (eg USA - Guatemala, Brazil; WWF - Costa Rica, Philippines)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 55-57; AQA spec 3.1.2.3.4.	
05.1	• Very small gene pool so individuals are closely related and inbreeding is likely • Harmful recessive genes are expressed when both parents carry them: some goslings had thin hair-like feathers with poor insulation so survival in cold weather was low • A stud book records family trees so breeding is arranged between the least related individuals (often coordinated by a specialist zoo) • Exchange of individuals / semen between zoos; avoid hybridisation with related species / subspecies	4
05.2	<i>Any 3 from:</i> • Breeding is triggered by stimuli (day length, light, temperature, food, body fat) which may be unknown / not provided • Population interactions: mixing breeding and non-breeding individuals causes conflict; some need a choice of partners, others pair for life • Breeding habitat needed: courtship display sites, nest sites, social groups (flamingos only breed in large groups - mirrors used) • Stress / disturbance	3
05.3	• AI: semen collected (and can be stored / cryopreserved for years) and inserted into the female, so animals need not be transported or meet; avoids dangerous mating (elephants, rhinos) and partner rejection • Embryo transfer: a female of the endangered species (eg bongo) is treated with hormones to release many eggs, which are fertilised and implanted into surrogate females of a common related species (eland) • So many offspring are produced per cycle instead of one calf per year, overcoming the shortage of breeding females	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 33-35; AQA spec 3.1.2.3.3.	

Question	Marking guidance	Marks
06.1	• Selective breeding: parents chosen for desirable genetic characteristics (Limousin muscle growth, Merino wool, Cheviot hardiness); combines genes but characteristics of offspring not fully predictable; many generations needed; risk of inbreeding • Asexual reproduction (cuttings, runners, tissue culture, cloning): offspring genetically identical to the parent so characteristics are predictable and survival high; but no variation so no improvement and vulnerability to the same disease; cloning of animals (nucleus transfer into an enucleated egg, surrogate) still developing • Cross-breeding: two breeds combined to give a mix of characteristics with hybrid vigour (heterosis) and lower inbreeding risk, eg Zebu (heat tolerant) x Ayrshire (high milk yield)	6
06.2	• AI: semen from a male with desirable characteristics can fertilise many more females than natural mating; frozen semen stored and transported easily and used after the male dies • Embryo transfer: FSH stimulates a superior female to release many eggs, which are fertilised (IVF) with sperm from a superior male • Each embryo is implanted in a different surrogate female • So the donor produces many offspring per year rather than one per pregnancy	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-323; AQA spec 3.5.1.2.	
07.1	• Coordinating global data on biodiversity conservation (Red List) • Increasing understanding of the importance of biodiversity • Deploying nature-based solutions to global challenges in climate, food and sustainable development	3
07.2	• Decline = $(12\,000 - 2100) / 12\,000 \times 100 = 82.5\%$ • Decline over 80% meets the Critically Endangered criterion even though habitat area (400 km^2) and adults (2100) only meet Endangered • A species is placed in the highest category for which it meets any one criterion, so Critically Endangered	3
07.3	• Categorising a species requires a great deal of research (population, range, trends) and resources; about 85 000 of millions of species assessed • Key species in habitats under particular threat (eg lemurs indicating the state of Madagascar's forests) • Evolutionarily unique species / EDGE species; endemic species; keystone species; flagship species; species with fragmented populations	3
07.4	• Iberian lynx: Critically Endangered to Endangered after captive breeding and protection of two habitat areas (population trebled in 15 years); Fregate Island beetle after rat eradication	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-27; AQA spec 3.1.2.3.1.	

Question	Marking guidance	Marks
08.1	• Size: shows whether the population is viable / how urgent action is; sets a baseline to judge success; needed to estimate carrying capacity and sustainable off-take • Distribution: shows which habitats / areas are used and whether the population is fragmented into non-viable isolated groups needing corridors; identifies key areas to protect • Survival rate (mortality): identifies the life stage and causes of death (predation, disease, poaching) so management targets the limiting factor • Age structure: many young indicates a growing population; few young / ageing population indicates breeding failure or high juvenile mortality even if adults are numerous; k-selected species need breeding adults protected	8
08.2	• Direct counts from vehicles / aircraft / drones along transects, or camera traps with individual recognition • Sub-sample counts in known areas scaled up to the whole reserve; mark-release-recapture (Lincoln index) for smaller species	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-402; AQA spec 3.1.2.3.5.	
09.1	• The readings are precise but may not be accurate	1
09.2	• A result that differs from other results with which it was expected to be similar • Repeat the study / measurement: if it recurs it is a 'real' result caused by another variable; if not it was an error and can be excluded	2
09.3	• To test the reliability of the method / equipment • To decide the sample size, number of samples, spacing or timing needed to detect the variation without collecting unnecessary data	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-391; AQA spec 3.7.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Succession: pioneers, seres (hydrosere, lithosere, psammosere), modification of abiotic conditions, climax communities controlled by climate; deflected succession and plagioclimaxes • Application: deciding whether to manage (plagioclimax habitats need grazing, mowing, burning, coppicing, pollarding continued) or leave alone (climax habitats); habitat creation planning the stages; controlling scrub / secondary succession; understanding that new habitats change over time • Population dynamics: birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles, r- and k-selection, MSY • Application: setting sustainable harvest levels (fisheries, whaling, forestry); recognising k-selected species need strict protection and recover slowly; deciding when culling is needed (deer without wolves, introduced species) or when releases / captive breeding are needed (low breeding rate); predicting effects of reintroducing predators • Monitoring population size, distribution, survival rate and age structure to assess success; adaptation and range of tolerance guiding abiotic management • Evaluation: uncertainties (natural fluctuations, unknown carrying capacity); examples (Yellowstone wolves, Scottish deer, chalk grassland); judgement	9
10.2	• Any taxonomic group of organisms classified together by similarity (eg species, genus, family)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-85; AQA spec 3.1.3.	

Question	Marking guidance	Marks
11.1	Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Energy flow: autotrophs and heterotrophs; losses by reflection, transmission, respiration; 100 000 to 1 000 units; few food chains beyond four levels; plant food most productive; grazing animals where crops impossible; pigs on wastes • Energetics: intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • Control of genetics: asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • GM: Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape; advantages; risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity) • Sustainability links: energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	25

Question	Marking guidance	Marks
11.2	Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Sustainability definitions; population and per capita impact; attitudes of rural and urban societies • Negative feedback: cloud cover and albedo, CO₂ and photosynthesis, evaporation and precipitation, density-dependent population regulation • Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition • Tipping points: rare under negative feedback; large events (Permian methane hydrate); self-sustaining change from human activities • Diversity and resistance: reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators • Human responses that override feedback (fertilisers, irrigation, pesticides); links to climate change chapter; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	25